2026-1 Data Structure

End-term Project

◎ Due date

6/6 (Sat) 23:59

Late submission will be penalized.

◎ Submission

LearnUs

◎ Caution

- Please upload your codes(*.cpp), report(*.pdf), and presentation video(*.mp4) **(less than 5min)**
  with a file name of **Project_NAME_ID.zip** at LearnUs
- Reports can be written both in **Kor. or Eng.**, and there is no fixed format.
- Explain your implementation and discuss your results.

[Project]

Select and use a suitable data structure for the type of the data and implement various functions for users to offer useful services(e.g. management system, or whatever you want)

☑ Term Project

☑ Personal Project

☑ Use C++

The project proceeds as follows.

1. Data Acquisition
   - Collect data to be used. Accumulate data through a survey, filming, recording, etc.
   - Data is selected as a type that can be used in a variety of ways.
   - **It is recommended to collect more than an appropriate amount to produce meaningful results.**
   - If you find them hard to collect, we recommend utilizing the following public data portals for your data collection (Again, no need to use the whole data. It needs to be *appropriate*):

   https://www.data.go.kr/index.do

   https://kosis.kr/index/index.do

   https://dataon.kisti.re.kr/
2. Data Storage
   - To efficiently store the acquired data, you can select the most suitable data structure among those covered or even uncovered during the semester (Queue, Stack, Tree, Graph, Hash, etc.) based on your own judgment.
   - The selection must be according to an individual's judgment so that it can be used well in the implemented functions.
3. Function Realization
   - Implement the functions for the basic operation appropriate for each data structure.
   - Consider various functions to utilize the data.
   - Based on the collected data, implement various functions to offer useful information.

[Project idea example]

A simple example of the project is attached. In your project, it is recommended to implement various functions based on the acquired data.

[Example 1] Subway line 2

- Data Acquisition: Collect the time required between stations and the route information of Seoul Subway Line 2.
- Data Storage: In the case of Line 2, a circular type with the outer ring and inner ring line, the Doubly Circular Linked List structure is used.
- Functions: Implement basic functions to access the previous node and next node in the same way as Doubly Linked List.
- Implement additional functions that inform the minimum required time, the direction of the train, and the number of passing stations between the two stations. Furthermore, implement the function that shows which stations are passed by between the two stations.

[Example 2] Ticket management service

- Data Acquisition: Collect customer service inquiry tickets, including reception time, issue category (e.g., simple question, payment error, server crash) and customer membership tier.
- Data Storage: Because urgent issues or inquiries from VIP customers must be handled first regardless of the order they arrived, a Priority Queue structure is used instead of a standard Queue.
- Functions: Implement basic functions to enqueue new tickets and dequeue the ticket with the highest priority.
- Implement additional functions that calculate the average waiting time per category and provide daily operation statistics. Furthermore, implement a function to temporarily hold a ticket and move it to a Deque if it requires further investigation.

Your **Project_NAME_ID.zip** must consist of..

1. Project Report

Submit a detailed report covering the entire project process.

The report must include the rationale for choosing your topic, the justification for the selected data structure, and the operating principles of your core functions.

Explain WHERE and HOW the data structures learned in class (Queue, Stack, Tree, etc.) were used in your program, and WHY you chose that specific structure among many others (proving its validity and efficiency).

Both Korean and English allowed.

2. Your Code

3. Presentation Video (approx. 5 minutes)

Record and submit a 5-minute video presenting and explaining your project.

The video must at lest contain explanation about your subject, code, and why you chose to do so, and your implementation results.